Omar Dennaoui

Department of Mathematics
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Born: August 21, 1999
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Education

University of Louisiana at Lafayette (ULL), Lafayette, LA, USA
- Ph.D. Mathematics — Jan 2021 –
- Dissertation: *TBA*
- Supervisors: Justin Lynd and Philip Hackney

University of Balamand, Balamand, El-Koura, Lebanon
- B.S. Mathematics, with honors — Jan 2017 – Jun 2020

Positions

University of Louisiana at Lafayette
- Graduate Teaching Assistant — Jan 2021 – Dec 2024
- Graduate Research Assistant (Dr. Philip Hackney) — Jan 2025 – May 2026

Research Interests

Structure and representation theory of finite groups, group cohomology. Fusion systems and related homotopy theory and homological algebra.
Higher Segal conditions in partial groups, localities, and higher categorical structures; multicategories and cyclic operads.

Publications

Contractibility of the orbit space of a saturated fusion system after Steinberg (doi, arXiv, pdf), with Jonathon Villareal J. Algebra **677** (2025), 267-277.

In progress

Monads and nerve theorems of higher categorical structures, 11pp.

Grants & Awards

Dissertation Completion Fellowship (full tuition waiver, stipend), ULL	FA 2026 – SP 2027
Graduate Research Assistantship (full tuition waiver, stipend), ULL	SP 2025 – SP 2026
Honors Convocation and Academic Excellence Award, ULL	SP 2024
Graduate Teaching Assistantship (full tuition waiver, stipend), ULL	SP 2021 – FA 2024

Talks

Contributed

Representations of Finite Groups and Related Structures — August 2026
University of California, Santa Cruz
Higher Segal Conditions and Localities of Symmetric and Alternating Groups at the Prime 2

The International Category Theory Conference & Discoveries and Vistas — July 2026
Johns Hopkins University
Higher Segal Conditions and Localities of Symmetric and Alternating Groups at the Prime 2

2026 Southern Regional Algebra Conference March 2026
Southeastern Louisiana University
Higher Segal Conditions and Localities of Symmetric and Alternating Groups at the Prime 2

2025 Southern Regional Algebra Conference in conjunction with Lloyd Roeling UL Lafayette Mathematics Conference April 2025
University of Louisiana at Lafayette
Contractibility of the orbit space of a saturated fusion system after Steinberg

The South Central Topology Conference IV March 2025
Texas A&M University
Contractibility of the orbit space of a saturated fusion system after Steinberg

Zassenhaus Groups and Friends Conference June 2024
Texas State University
Contractibility of the orbit space of a saturated fusion system after Steinberg

The South Central Topology Conference III October 2023
New Mexico State University
Contractibility of the orbit space of a saturated fusion system after Steinberg

Local & Expository

Algebra Seminar, University of Louisiana at Lafayette November 2023
Contractibility of the orbit space of a saturated fusion system after Steinberg

Algebra Seminar, University of Louisiana at Lafayette March 2022
Limits and colimits, generators and relations of partial groups

Teaching Experience

Courses Taught at UL Lafayette

MTHS 105/105S: Applied College Algebra Fundamentals	FA 2024
MTHS 109/109S: Pre-Calculus Algebra	SP 2024
MTHS 105/105S: Applied College Algebra Fundamentals	FA 2023
MATH 105: Applied College Algebra	SP 2023
MATH 103/104: Applied College Algebra Fundamentals	FA 2022

University and Departmental

President, UL Graduate Student Chapter of The American Mathematical Society	FA 2023 – SP 2024
Secretary, UL Graduate Student Chapter of The American Mathematical Society	FA 2022 – SP 2023
Member, UL Graduate Student Chapter of The American Mathematical Society	FA 2021 –

Other Professional Activity

Member, Mathematical Association of America	2021 –
Member, The American Mathematical Society	2021 –